| Field | Entry |
|---|---|
| **Source ID** | WBC-0004 |
| **Title** | The Mechanics of Splitting Wood and the Design of Neolithic Woodworking Tools |
| **Source type** | Paper (peer-reviewed, open access) |
| **Link / File** | https://exarc.net/issue-2017-4/ea/mechanics-splitting-wood-and-design-neolithic-woodworking-tools |
| **Craft applicability** | Wood (handle/socket construction — applies to wood and bamboo tool handles) |
| **Primary Why-theme** | 10. Failure, Repair and Quality (with a secondary link to 7. Tools and tool-material interaction) |
| **Process stage** | Tool maintenance / primary shaping (tool condition affecting all carving stages) |
| **Questions answered / generated** | Why does the tool handle split near the socket under repeated impact? Why does a metal ring or flange at the socket reduce this failure? |
| **Evidence level** | E3 (direct scientific evidence, experimentally validated model) |
| **Relevance note** | Explains the mechanics of radial wood splitting at handle/socket junctions under impact and shows why reinforcing collars/flanges at the socket reduce failure — directly supporting the artisan practice of fitting an iron ring to the neem handle of the *baria*. |